

WEEK 2 – POWER BI REPORT

SUPERSTORE DATA PREPARATION

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The Superstore Sales dataset was successfully imported into Power BI using Power Query. The samplesuperstore table serves as the primary fact table and contains the original sales transaction records used for developing the analytical dashboard.

The screenshot displays the Power BI Power Query Editor interface. The main area shows a data preview of the 'samplesuperstore' table, which contains 28 columns and 999+ rows. The columns are: Order ID, Order Date, Ship Date, Ship Mode, Customer ID, and Customer Name. The data is filtered to show the top 1000 rows. A context menu is open over the 'Order ID' column, showing options like Copy, Paste, Delete, Rename, and Advanced Editor. The 'Advanced Editor' tab is selected, showing the formula: `Table.RenameColumns(F"Inserted Day Name",{"Day Name", "Order Day of Week"})`. The 'Query Settings' pane on the right shows the query name 'samplesuperstore' and a list of applied steps: Source, Promoted Headers, Changed Type, Changed Type with Locale, Changed Type1, Trimmed Text, Cleaned Text, Trimmed Text1, Cleaned Text1, Inserted Year, Renamed Columns, Inserted Quarter, Renamed Columns1, Added Custom, Inserted Month, Renamed Columns2, Inserted Month Name, Renamed Columns3, Inserted Day, Renamed Columns4, Inserted Day Name, and Renamed Columns5. The status bar at the bottom indicates '28 COLUMNS, 999+ ROWS' and 'Column profiling based on top 1000 rows'.

Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name
1	US-2023-103800	03-01-2023	07-01-2023 Standard Class	DP-13000	Darren Powers
2	US-2023-112326	04-01-2023	08-01-2023 Standard Class	PO-19195	Phyllina Ober
3	US-2023-112326	04-01-2023	08-01-2023 Standard Class	PO-19195	Phyllina Ober
4	US-2023-112326	04-01-2023	08-01-2023 Standard Class	PO-19195	Phyllina Ober
5	US-2023-141817	05-01-2023	12-01-2023 Standard Class	MB-18085	Mick Brown
6	US-2023-167199	06-01-2023	10-01-2023 Standard Class	ME-17320	Maria Ehszadi
7	US-2023-167199	06-01-2023	10-01-2023 Standard Class	ME-17320	Maria Ehszadi
8	US-2023-106054	06-01-2023	07-01-2023 First Class	JO-15145	Jack O'Brian
9	US-2023-167199	06-01-2023	10-01-2023 Standard Class	ME-17320	Maria Ehszadi
10	US-2023-167199	06-01-2023	10-01-2023 Standard Class	ME-17320	Maria Ehszadi
11	US-2023-167199	06-01-2023	10-01-2023 Standard Class	ME-17320	Maria Ehszadi
12	US-2023-130813	06-01-2023	08-01-2023 Second Class	LS-17230	Lyornis Saunders
13	US-2023-167199	06-01-2023	10-01-2023 Standard Class	ME-17320	Maria Ehszadi
14	US-2023-167199	06-01-2023	10-01-2023 Standard Class	ME-17320	Maria Ehszadi
15	US-2023-105417	07-01-2023	12-01-2023 Standard Class	VS-21820	Vivek Sundaresam
16	US-2023-105417	07-01-2023	12-01-2023 Standard Class	VS-21820	Vivek Sundaresam
17	US-2023-135405	09-01-2023	13-01-2023 Standard Class	MS-17830	Melanie Sette
18	US-2023-135405	09-01-2023	13-01-2023 Standard Class	MS-17830	Melanie Sette
19	US-2023-149020	10-01-2023	15-01-2023 Standard Class	AI-10780	Anthony Jacobs
20	US-2023-149020	10-01-2023	15-01-2023 Standard Class	AI-10780	Anthony Jacobs
21	US-2023-130092	11-01-2023	14-01-2023 First Class	SV-20365	Seth Vernon
22	US-2023-157147	13-01-2023	16-01-2023 Standard Class	BD-11605	Brian Dahlen
23	US-2023-109232	13-01-2023	16-01-2023 Second Class	ND-18370	Natalie DeCherney
24	US-2023-157147	13-01-2023	16-01-2023 Standard Class	BD-11605	Brian Dahlen

A reference of the samplesuperstore table was created in Power Query to facilitate the development of separate dimension tables. This approach helps structure the dataset efficiently and establishes a well-organized data model for analytical reporting.

The screenshot shows the Tableau Desktop interface with the 'saalesuperstore' table loaded. A right-click context menu is open over the 'Customer ID' column, showing options like Copy, Paste, Delete, Rename, Enable load, Include in report refresh, Duplicate, Reference, Move To Group, Move Up, Move Down, Create Function..., Convert To Parameter, Advanced Editor, and Properties... The table columns include Order ID, Order Date, Ship Date, Ship Mode, Customer ID, and Customer Name. The status bar at the bottom indicates '28 COLUMNS, 999+ ROWS' and 'Column profiling based on top 1000 rows'.

A dedicated Customers dimension table was created from the Superstore dataset by selecting relevant customer attributes. The table includes key fields such as Customer ID, Customer Name, Segment, City, Country/Region, Postal Code, and Region, enabling detailed customer-level analysis.

The screenshot shows the Tableau Desktop interface with the 'Customers' table loaded. A right-click context menu is open over the 'Segment' column, showing options like Copy, Remove Columns, Remove Other Columns, Add Column From Examples..., Remove Duplicates, Remove Errors, Replace Values..., Fill, Change Type, Transform, Merge Columns, Group By..., Unpivot Columns, Unpivot Other Columns, and Unpivot Only Selected Columns. The table columns include Order ID, Order Date, Ship Date, Ship Mode, Customer ID, and Segment. The status bar at the bottom indicates '28 COLUMNS, 999+ ROWS' and 'Column profiling based on top 1000 rows'.

Irrelevant and redundant fields were removed from the referenced customer table, retaining only the attributes required for analysis. This data-cleaning step results in a streamlined Customers dimension that supports efficient customer segmentation and reporting.

Table: SelectColumns(Source, {"Customer ID", "Customer Name", "Segment", "Country/Region", "City", "Postal Code", "Region", "State/Province"})

Customer Name	Segment	Country/Region	City	Postal Code	Region	State/Province
1. Neil Powers	Consumer	United States	Houston	77095	Central	Texas
2. Nina Ober	Home Office	United States	Naperville	60540	Central	Illinois
3. Nina Ober	Home Office	United States	Naperville	60540	Central	Illinois
4. Nina Ober	Home Office	United States	Naperville	60540	Central	Illinois
5. J. Brown	Consumer	United States	Philadelphia	19143	East	Pennsylvania
6. Nina Etzadi	Home Office	United States	Henderson	42420	South	Kentucky
7. Nina Etzadi	Home Office	United States	Henderson	42420	South	Kentucky
8. J. O'Brant	Corporate	United States	Athens	30605	South	Georgia
9. Nina Etzadi	Home Office	United States	Henderson	42420	South	Kentucky
10. Nina Etzadi	Home Office	United States	Henderson	42420	South	Kentucky
11. Nina Etzadi	Home Office	United States	Henderson	42420	South	Kentucky
12. J. Saunders	Consumer	United States	Los Angeles	90049	West	California
13. Nina Etzadi	Home Office	United States	Henderson	42420	South	Kentucky
14. Nina Etzadi	Home Office	United States	Henderson	42420	South	Kentucky
15. J. Sundaresam	Consumer	United States	Huntsville	77340	Central	Texas
16. J. Sundaresam	Consumer	United States	Huntsville	77340	Central	Texas
17. J. Saito	Consumer	United States	Laredo	78041	Central	Texas
18. J. Saito	Consumer	United States	Laredo	78041	Central	Texas
19. J. Jacobs	Corporate	United States	Springfield	22153	South	Virginia
20. J. Jacobs	Corporate	United States	Springfield	22153	South	Virginia
21. J. Varnon	Consumer	United States	Dover	19001	East	Delaware
22. J. Dahlin	Consumer	United States	San Francisco	94109	West	California
23. J. DeCherney	Consumer	United States	Mount Pleasant	29664	South	South Carolina
24. J. Dahlin	Consumer	United States	San Francisco	94109	West	California
25. J. Dahlin	Consumer	United States	San Francisco	94109	West	California

A dedicated Products dimension table was created using relevant product attributes from the Superstore dataset. The table contains Product ID, Category, Sub-Category, and Product Name, providing a structured foundation for product-level analysis.

Table: SelectColumns(Source, {"Product ID", "Category", "Sub-Category", "Product Name"})

Product ID	Category	Sub-Category	Product Name
1. OFF-PA-10000174	Office Supplies	Paper	Message Rock, Wirebound, Four 5 1/2" x 4" Forms/Pg., 200 Dapt. Set...
2. OFF-BI-10004094	Office Supplies	Binders	QBC Standard Plastic Binding Systems Combo
3. OFF-LA-10003223	Office Supplies	Labels	Avery 508
4. OFF-ST-10002743	Office Supplies	Storage	SAFCD Boltless Steel Shelving
5. OFF-AR-10003478	Office Supplies	Art	Avery 91-Liter EverBold Pen Style Fluorescent Highlighters, 4/Pack
6. FUR-CH-10004063	Furniture	Chairs	Global Deluxe High-Back Manager's Chair
7. OFF-AR-10001862	Office Supplies	Art	Rogers Handheld Barrel Pencil Sharpener
8. OFF-AR-10002399	Office Supplies	Art	Dixon Prang Watercolor Pencils, 10-Color Set with Brush
9. OFF-BI-10004632	Office Supplies	Binders	Itasca Hi-Tech Manual Binding System
10. OFF-FA-10001883	Office Supplies	Fasteners	Alliance Super-Size Bands, Assorted Sizes
11. OFF-PA-10000955	Office Supplies	Paper	Southworth 25% Cotton Granite Paper & Envelopes
12. OFF-PA-10002005	Office Supplies	Paper	Xerox 225
13. TEC-PH-10004539	Technology	Phones	Wireless Extenders zBoost V9345 SOHO Signal Booster
14. TEC-PH-10004377	Technology	Phones	GE 30524E4
15. FUR-FU-10004864	Furniture	Furnishings	Howard Miller 14-1/2" Diameter Chrome Round Wall Clock
16. OFF-BI-10003708	Office Supplies	Binders	Acco Four Pocket Poly Ring Binder with Label holder, Smoke, 1"
17. OFF-AR-10004078	Office Supplies	Art	Newell 312
18. TEC-AC-10001266	Technology	Accessories	Memorex Micro Travel Drive 8 GB
19. FUR-FU-10000965	Furniture	Furnishings	Howard Miller 11-1/2" Diameter Ridgewood Wall Clock
20. OFF-LA-10004272	Office Supplies	Labels	Avery 482
21. FUR-FU-10000010	Furniture	Furnishings	DAX Value U-Channel Document Frames, Esal Back
22. FUR-BO-10003034	Furniture	Bookcases	O'Sullivan Elevations Bookcase, Cherry Finish
23. FUR-CH-10000422	Furniture	Chairs	Global Highback Leather Tiller in Burgundy
24. OFF-AR-10003514	Office Supplies	Art	4009 Highlighters by Sanford
25. OFF-BI-10003476	Office Supplies	Binders	Avery Metallic Poly Binders

A dedicated Date dimension was created using DAX to support time-based analysis. The table dynamically generates dates ranging from the minimum to the maximum Order Date and includes additional attributes such as Day, Month, Quarter, and Year for effective trend and period analysis.

The screenshot shows the Power BI Desktop interface with a DAX query in the formula bar and a data preview below it.

DAX Query:

```

1 Din_Date =
2 ADDCOLUMNS(
3     CALENDAR(
4         MIN(samplestore[Order Date]),
5         MAX(samplestore[Order Date])
6     ),
7     "Day", DAY([Date]),
8     "Month", FORMAT([Date], "MMMM"),
9     "Quarter", "Q" & FORMAT([Date], "Q"),
10    "Year", YEAR([Date])
11 )

```

Data Preview:

Date	Day	Month	Quarter	Year
01-07-2023 00:00:00	1	July	Q3	2023
02-07-2023 00:00:00	2	July	Q3	2023
03-07-2023 00:00:00	3	July	Q3	2023
04-07-2023 00:00:00	4	July	Q3	2023
05-07-2023 00:00:00	5	July	Q3	2023
06-07-2023 00:00:00	6	July	Q3	2023
07-07-2023 00:00:00	7	July	Q3	2023
08-07-2023 00:00:00	8	July	Q3	2023
09-07-2023 00:00:00	9	July	Q3	2023
10-07-2023 00:00:00	10	July	Q3	2023
11-07-2023 00:00:00	11	July	Q3	2023
12-07-2023 00:00:00	12	July	Q3	2023
13-07-2023 00:00:00	13	July	Q3	2023
14-07-2023 00:00:00	14	July	Q3	2023
15-07-2023 00:00:00	15	July	Q3	2023
16-07-2023 00:00:00	16	July	Q3	2023
17-07-2023 00:00:00	17	July	Q3	2023
18-07-2023 00:00:00	18	July	Q3	2023

Table: Din_Date (1,658 rows)

Duplicate Customer IDs were removed from the Customers dimension to ensure that each customer is represented uniquely. This establishes Customer ID as a reliable key for creating relationships between the Customers dimension and the sales fact table.

The screenshot shows the Power BI Desktop interface with a DAX query in the formula bar and a data preview below it.

DAX Query:

```

1 Customer =
2 Table.Distinct(
3     RemoveOtherColumns(
4         Customers,
5         {"Customer ID"}
6     )
7 )

```

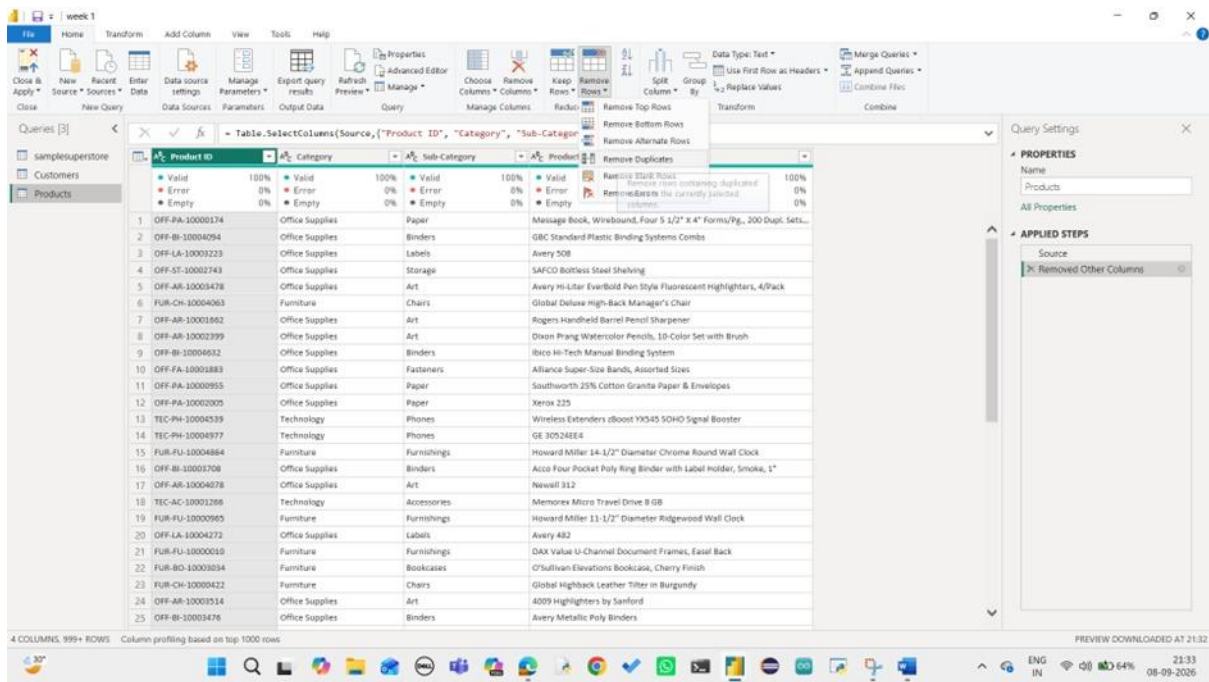
Data Preview:

Customer ID	Customer Name	Segment	Country	Postal Code	Region
DP-13000	Darren Powers	Consumer	United States	77095	Central
PO-19195	Phyllis Ober	Home Office	United States	02540	Central
MB-18085	Mick Brown	Consumer	United States	19143	East
ME-17320	Maria Elzadi	Home Office	United States	42420	South
JO-15145	Jack O'Brian	Corporate	United States	30605	South
LS-17230	Lynoric Saunders	Consumer	United States	90049	West
VY-21820	Viviek Sandersam	Consumer	United States	77340	Central
MS-17830	Melanie Sette	Consumer	United States	78041	Central
AI-10780	Anthony Jacobs	Corporate	United States	22153	South
SV-20365	Seth Vernon	Consumer	United States	19901	East
BD-11905	Brian Dahlen	Consumer	United States	94109	West
ND-18370	Natalie DeCherney	Consumer	United States	29404	South
MM-17920	Michael Moore	Consumer	United States	43055	East
CS-12250	Chris Selesnick	Corporate	United States	71111	South
BS-11590	Brendan Sward	Corporate	United States	19140	East
EH-13990	Erica Hackney	Consumer	United States	30076	South
DL-13315	Delfina Litchford	Consumer	United States	19134	East
DW-13135	David Wiener	Corporate	United States	97477	West
TS-21340	Toby Swindell	Consumer	United States	85254	West
HL-15040	Hunter Lopez	Consumer	United States	72401	South
XP-21865	Xylona Prais	Consumer	United States	48185	Central
MV-17485	Mark Van Huff	Consumer	United States	90049	West
MM-18280	Muhammed MacIntyre	Corporate	United States	37187	South
TP-21400	Tom Boeckenhauer	Consumer	United States	33180	South

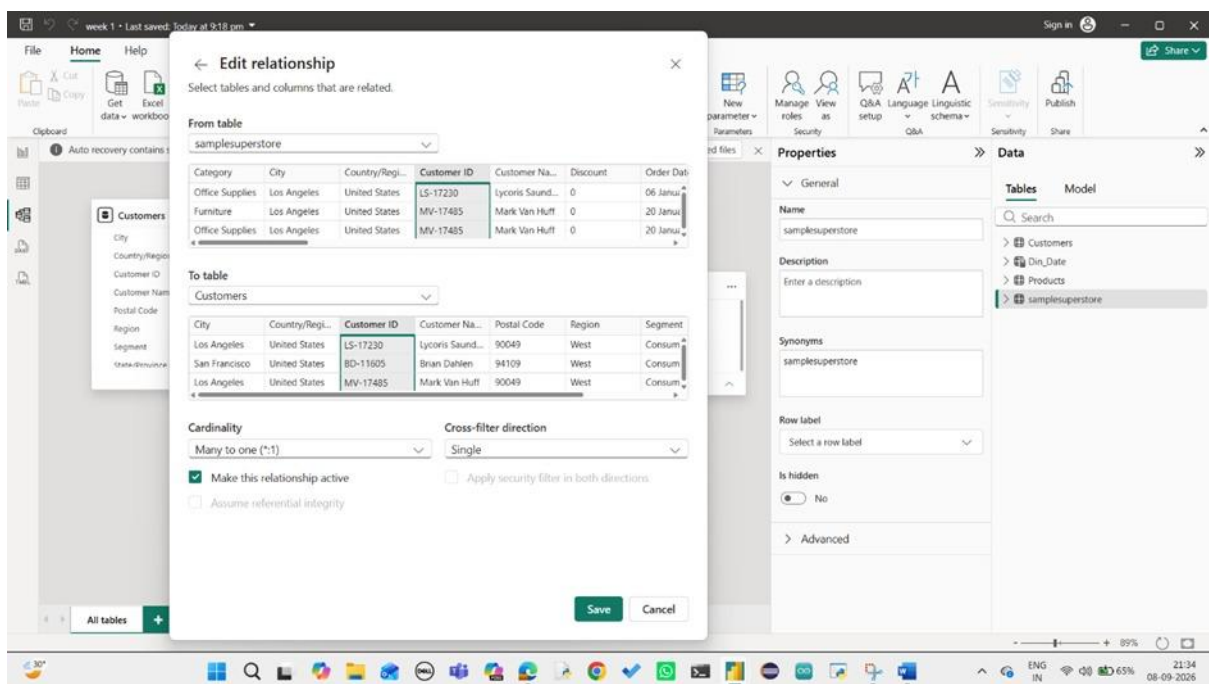
8 COLUMNS, 804 ROWS. Column profiling based on top 1000 rows.

Duplicate Product IDs were removed from the Products dimension to maintain a single unique record for each product. This ensures accurate relationships with the sales fact table and supports reliable product-level reporting.

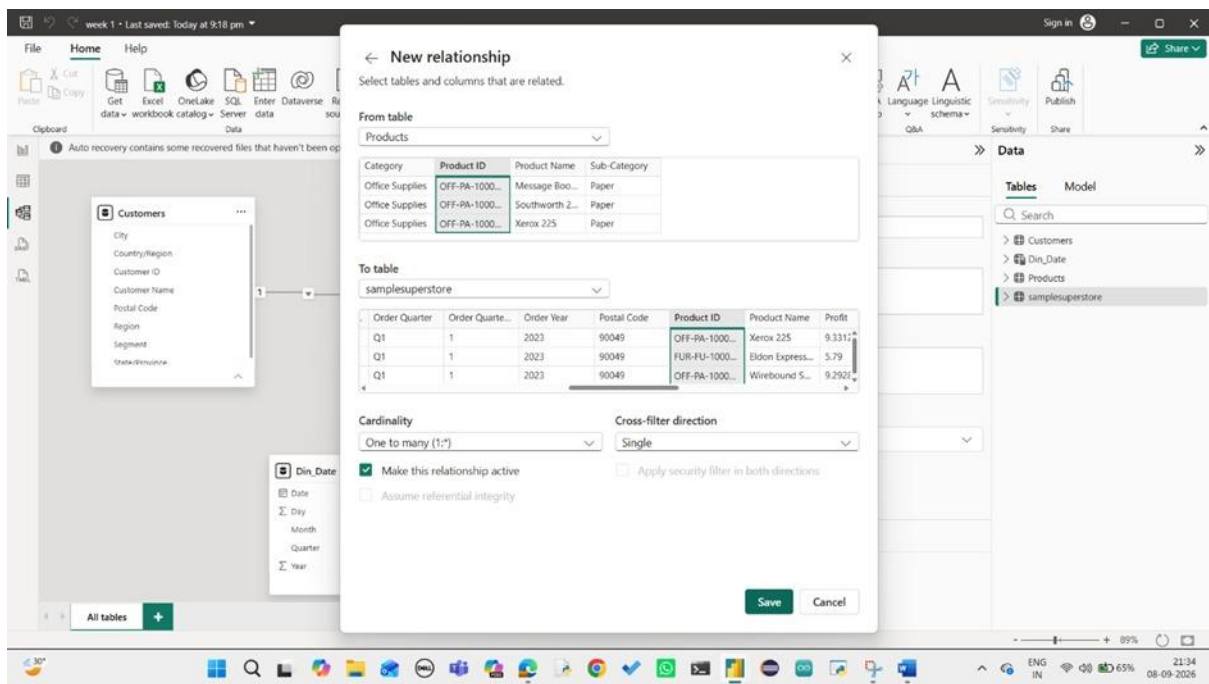
DATA MODEL



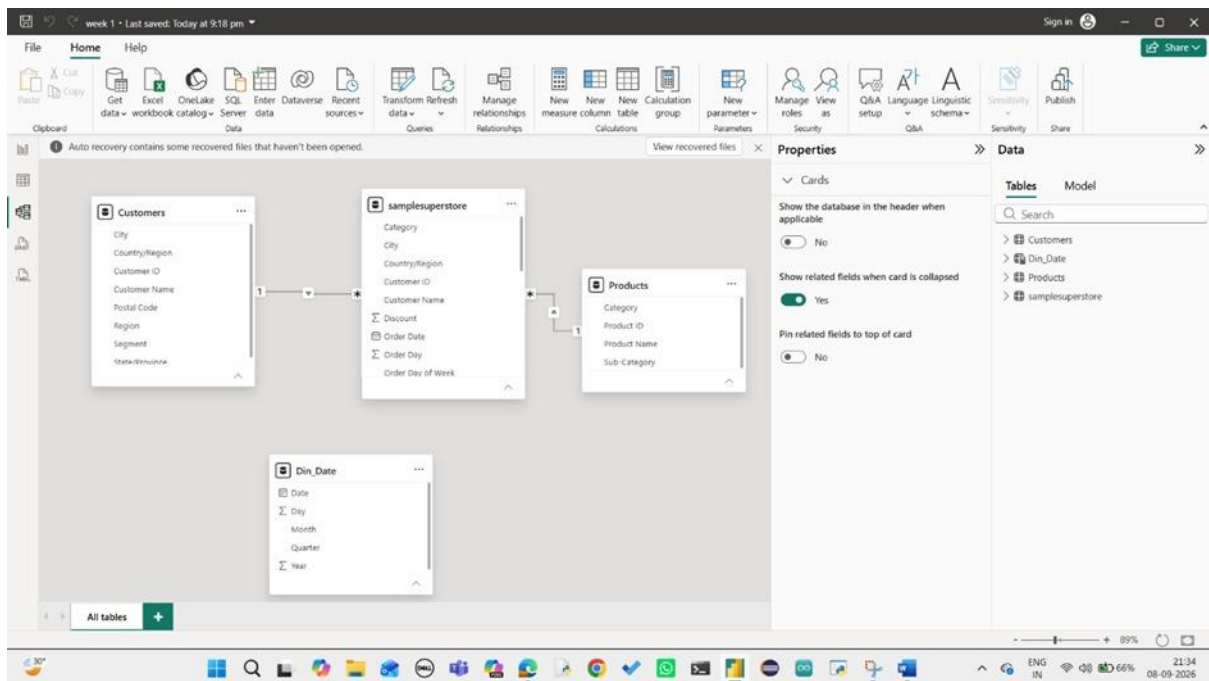
A relationship was established between the Customers dimension and the sample superstore fact table using Customer ID. This relationship connects customer attributes with their corresponding sales transactions, enabling customer-focused analysis across the report.



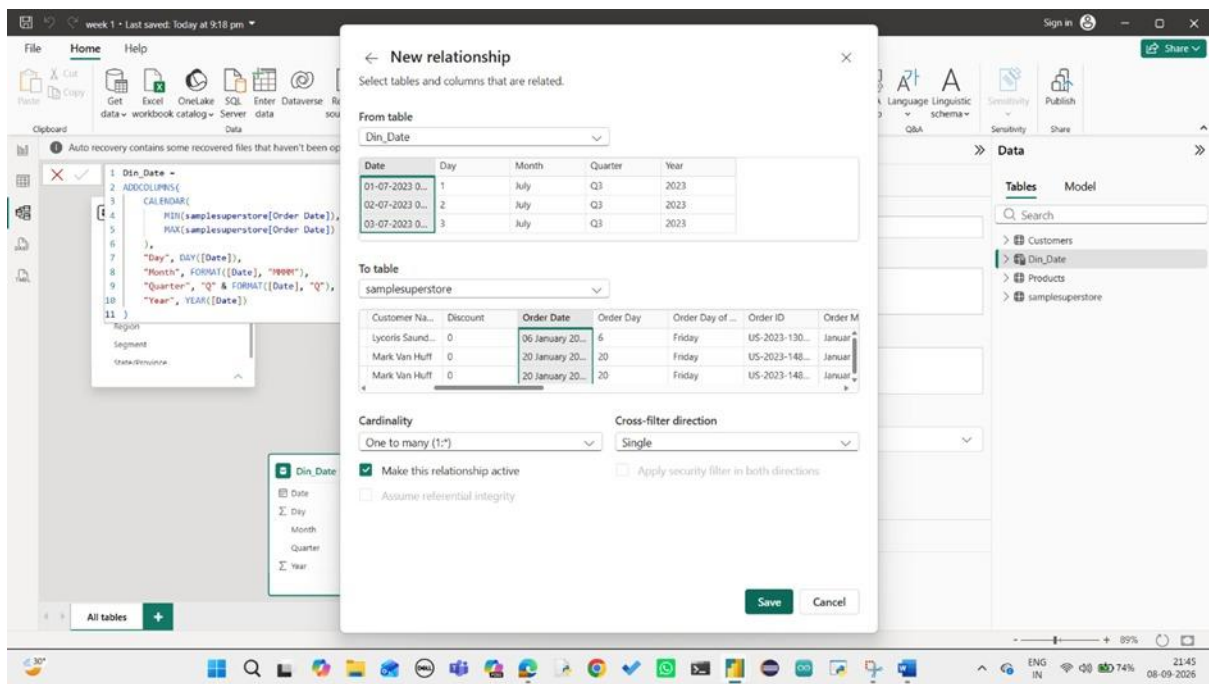
A relationship was established between the Products dimension and the samplesuperstore fact table using Product ID. This enables sales performance to be analyzed across product categories, sub-categories, and individual products.



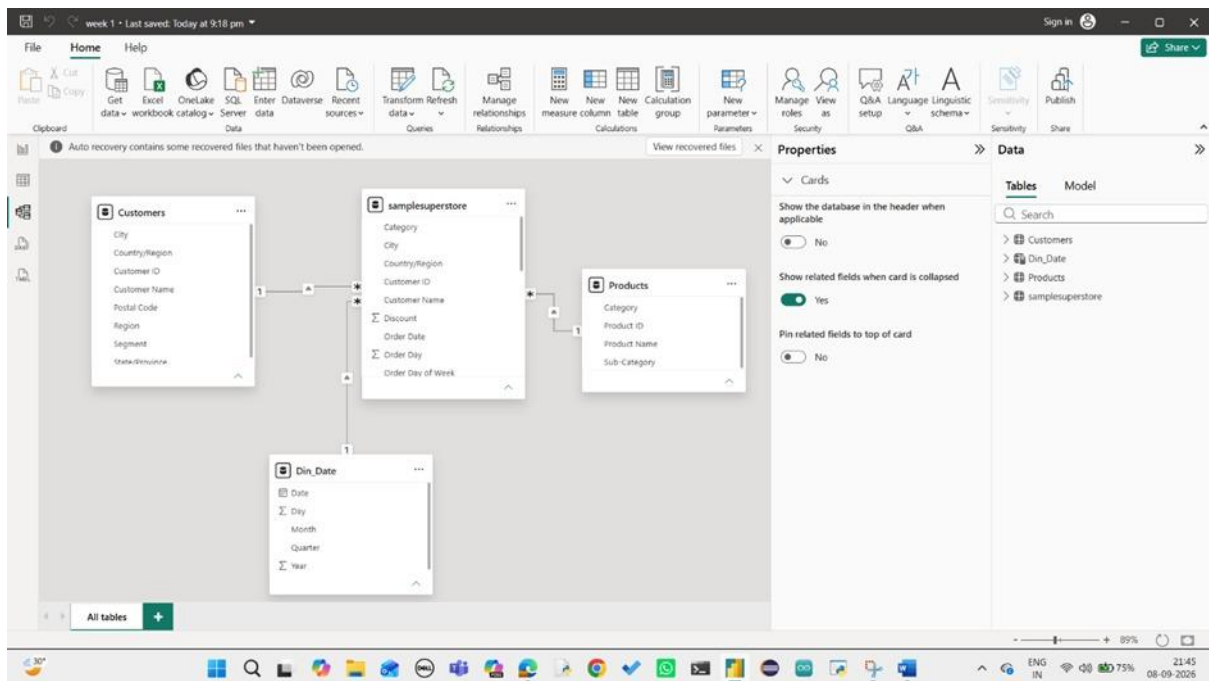
The Power BI data model follows a **Star Schema architecture**, with the samplesuperstore table serving as the central fact table. The Customers, Products, and Dim_Date tables function as supporting dimension tables connected to the fact table through well-defined relationships. This structure improves model organization, scalability, and analytical performance.



The Dim_Date table was connected to the samplesuperstore fact table using the Order Date field. A one-to-many relationship with single-directional filtering was implemented, allowing accurate and consistent time-based analysis across the dashboard.



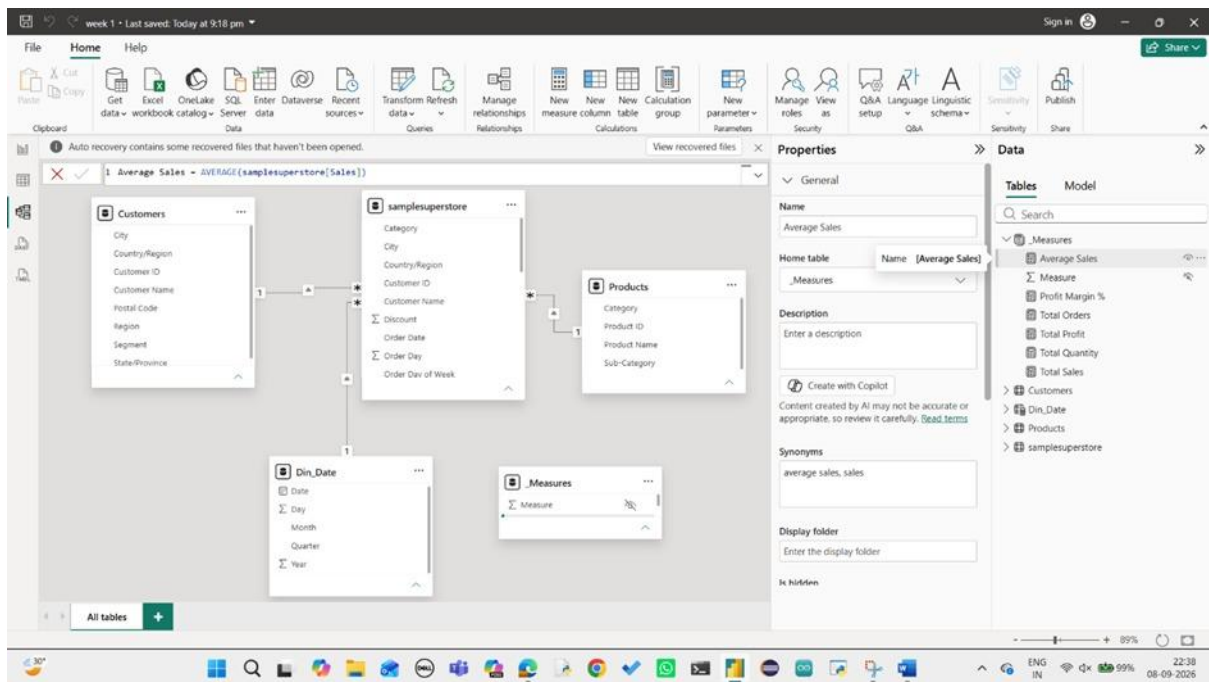
The completed data model integrates the central sales fact table with the supporting Customers, Products, and Date dimensions. The structured relationships provide a reliable foundation for interactive visualizations, filtering, and DAX-driven business analysis.



The Average Sales measure was created using the AVERAGE DAX function to determine the average sales value across the available sales transactions. This metric provides an overview of the typical transaction-level sales performance.

DAX:

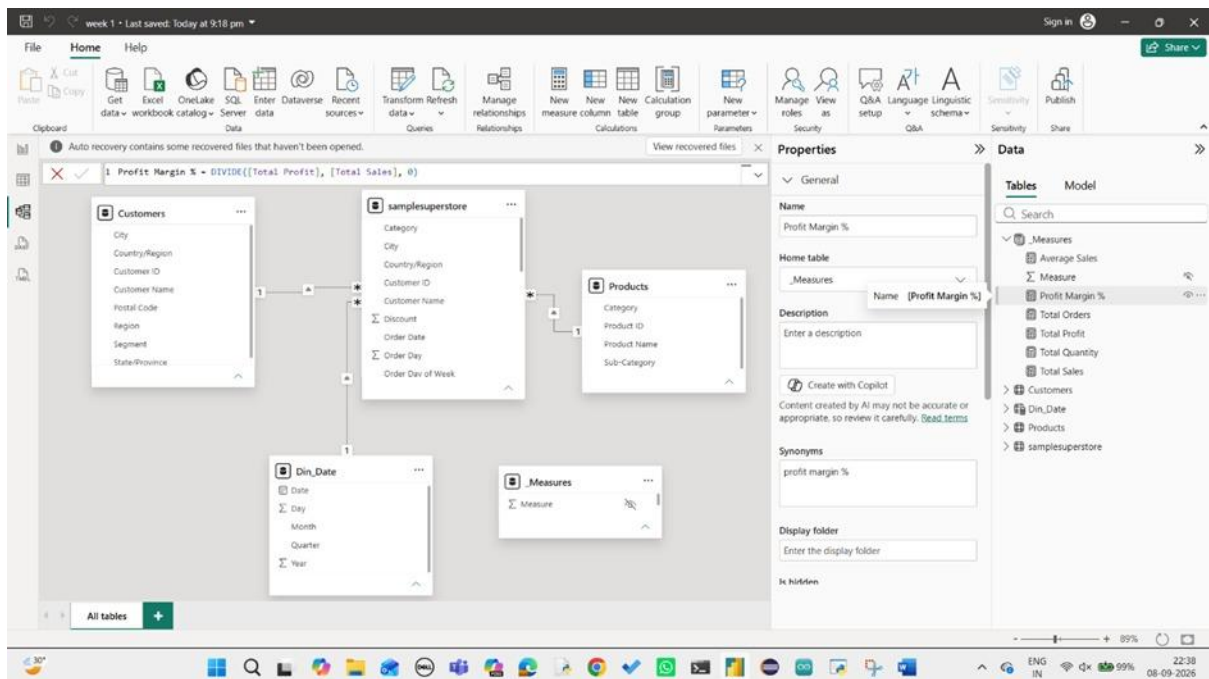
Average Sales = AVERAGE(samplesuperstore[Sales])



The Profit Margin % measure was developed to evaluate profitability relative to total sales. The DIVIDE function is used to perform the calculation safely while preventing errors in cases where total sales may be zero.

DAX:

Profit Margin % = DIVIDE([Total Profit], [Total Sales], 0)

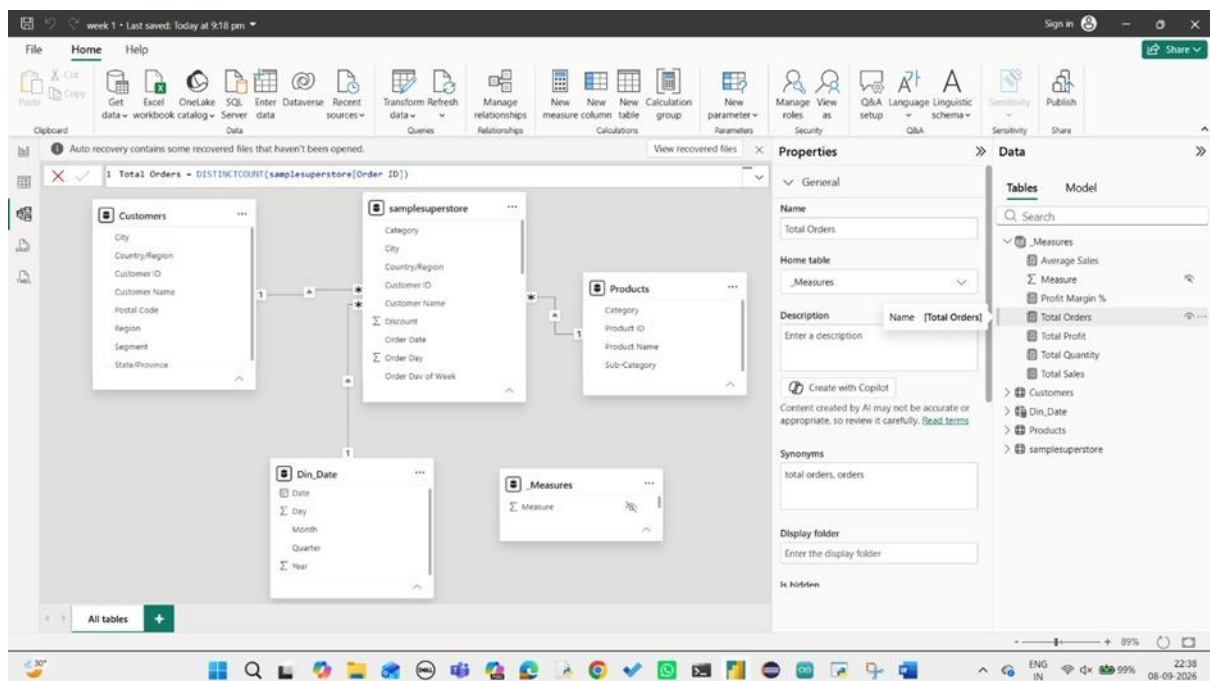


The Total Orders measure calculates the number of unique orders recorded in the Superstore dataset. DISTINCTCOUNT is applied to the Order ID field to

ensure that multiple transaction lines belonging to the same order are not counted as separate orders.

DAX:

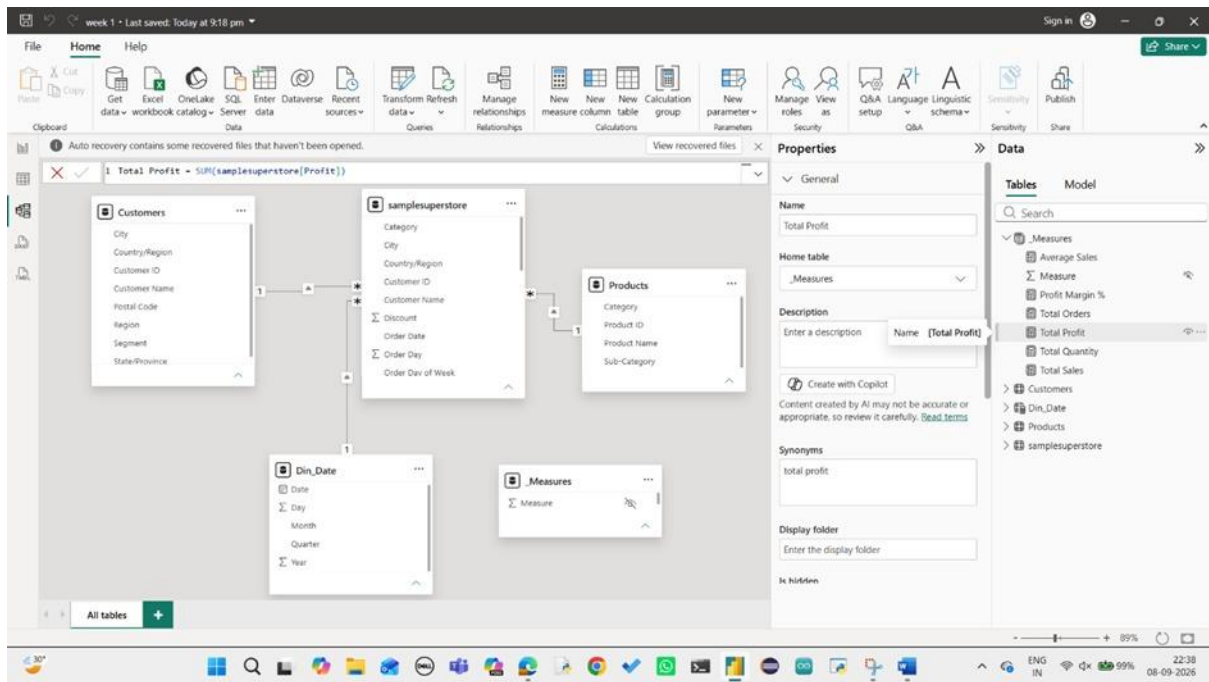
Total Orders = DISTINCTCOUNT(samplesuperstore[Order ID])



The Total Profit measure calculates the aggregate profit generated across all sales transactions. The SUM function is applied to the Profit column to provide a consolidated view of overall profitability.

DAX:

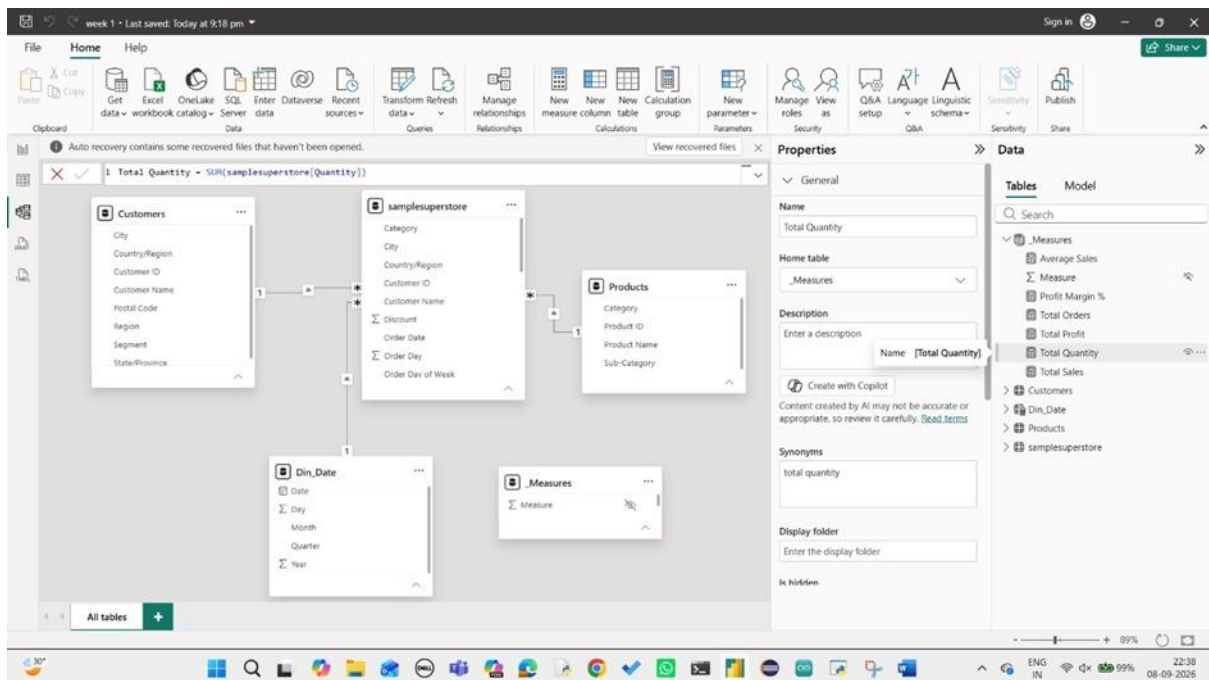
Total Profit = SUM(samplesuperstore[Profit])



The Total Quantity measure calculates the overall quantity of products sold across all recorded transactions. This metric provides insight into sales volume and product movement.

DAX:

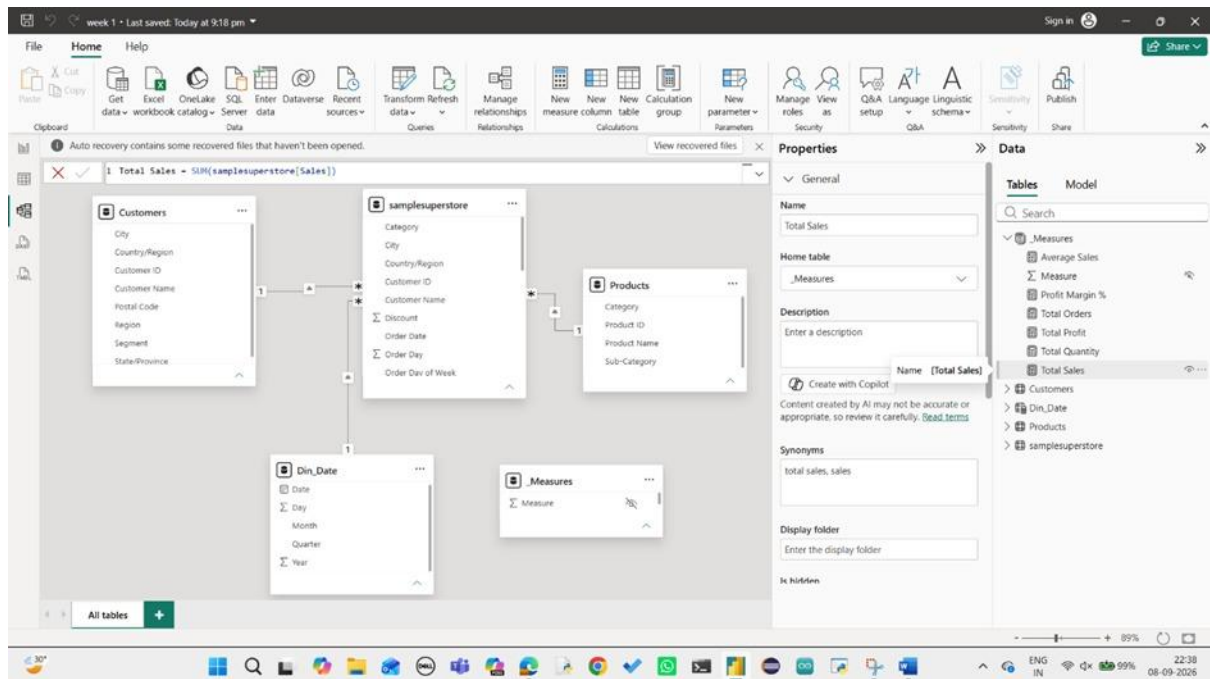
Total Quantity = SUM(samplesuperstore[Quantity])



The Total Sales measure calculates the overall sales revenue generated within the Superstore dataset. It serves as a key performance indicator and is utilized across multiple dashboard visuals, including KPI cards and comparative analyses.

DAX:

Total Sales = SUM(samplesuperstore[Sales])



The final Power BI dashboard delivers an interactive and comprehensive overview of Superstore sales performance. Key Performance Indicators (KPIs) highlight essential business metrics, including **Average Sales, Profit Margin, Total Orders, Total Profit, and Total Quantity**.

Interactive charts provide deeper insights into **sales trends, regional performance, and category-level performance**, enabling users to identify patterns and compare business results effectively. Slicers for **Year, Region, Category, and Segment** allow users to dynamically filter the report and explore the data from multiple business perspectives.

Overall, the dashboard combines **data transformation, dimensional modeling, DAX-based calculations, and interactive visualization** to create a structured and decision-oriented sales analytics solution.

